Embedded Systems Final Project

Industrial Motor Speed Controller

Student Name	Student ID
Haneen Awad	220212135

1. Introduction

This project implements an Industrial Motor Speed Controller using an Arduino UNO microcontroller, simulated in Proteus. The system allows the user to control a DC motor's speed via a potentiometer, display real-time RPM on a 20x4 LCD, and trigger an LED warning at high speed.

Motor speed control is essential in many industrial applications such as conveyors, fans, and pumps. PWM (Pulse Width Modulation) is a common and efficient technique for this purpose.

2. System Description

The system works as follows:

- On startup, the LCD shows the project title screen for 3 seconds.
- Then the LCD shows the student names and IDs for 3 seconds.
- In the main loop, the potentiometer value (0–1023) is read from pin A0.
- The value is mapped to a PWM signal (0–255) sent to pin D9 through R1 (220Ω) to the gate of the IRLZ44N MOSFET (Q1), which switches the 12V motor supply.
- The same value is mapped to a simulated RPM range (0–470) for display on the LCD.
- If RPM ≥ 230, the LED on pin D13 turns ON and the LCD shows 'High Speed'. Otherwise the LED is OFF.

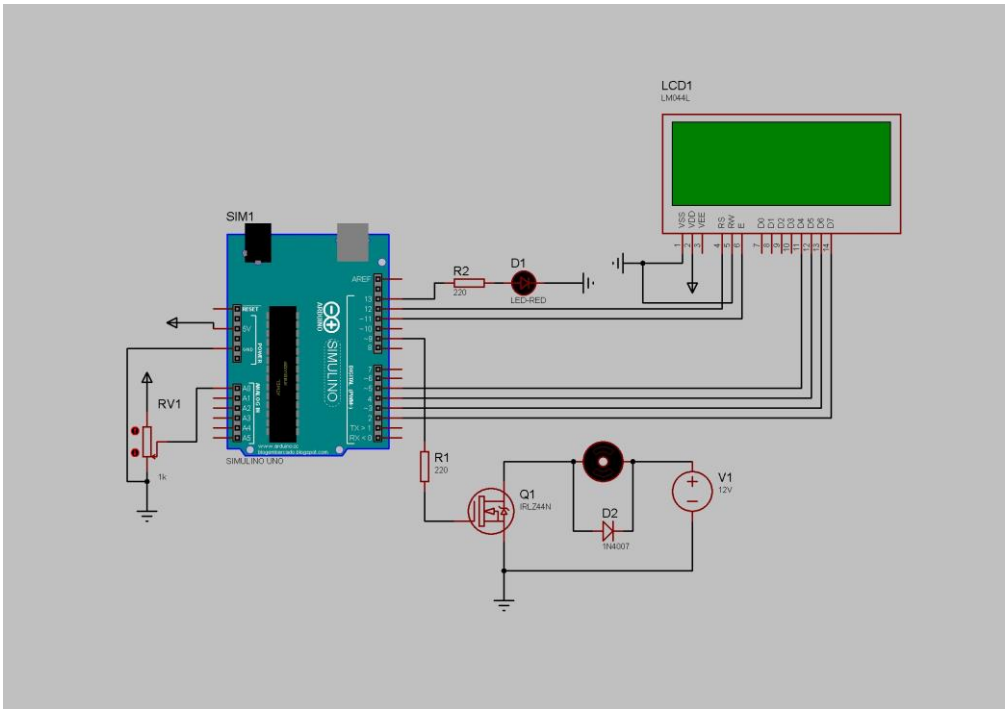
3. Components Used

#	Component	Qty	Role
1	Arduino UNO (SIM1)	1	Main microcontroller – reads sensors, generates PWM, drives LCD
2	Potentiometer (RV1 – 1kΩ)	1	Analog speed input (0–5V) connected to pin A0
3	DC Motor	1	Actuator – speed controlled by MOSFET PWM switching
4	N-ch MOSFET (Q1 – IRLZ44N)	1	Switches 12V motor current via PWM gate signal from D9

5	LCD 20x4 (LCD1 – LM044L)	1	Displays project info, student names, RPM and LED status
6	LED – Red (D1)	1	Visual warning indicator for high-speed operation (≥ 230 RPM)
7	Resistors R1, R2 – 220Ω each	2	R1: gate resistor for MOSFET; R2: current limiter for LED
8	Flyback Diode (D2 – 1N4007)	1	Protects MOSFET from motor back-EMF voltage spikes
9	DC Power Supply (V1 – 12V)	1	Dedicated 12V supply for the DC motor circuit

4. Circuit Diagram

The figure below shows the full Proteus simulation circuit. The potentiometer (RV1, 1kΩ) feeds into A0. Pin D9 generates a PWM signal through R1 (220Ω) to the gate of the IRLZ44N MOSFET (Q1), which switches the 12V motor supply. A 1N4007 flyback diode (D2) protects the MOSFET from back-EMF. The LM044L 20x4 LCD is connected in 4-bit mode, and the warning LED (D1) with series resistor R2 (220Ω) is on pin D13.



5. System Flowchart

The program follows this logical flow:

START / Power ON
Initialize LCD → Show Title Screen (3s) → Show Student Info (3s)
Read Potentiometer → A0 (0–1023)
Map → PWM (0–255) & RPM (0–470)
analogWrite(D9, PWM) → MOSFET → Drive Motor
DECISION: RPM ≥ 230?
YES → LED ON | LCD: 'LED: ON (High Speed)'
NO → LED OFF | LCD: 'LED: OFF (Low Speed)'
Wait 200ms → Repeat Loop

6. Arduino Code

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

const int potPin = A0;
const int motorPin = 9;
const int ledPin = 13;

void setup() {

  pinMode(motorPin, OUTPUT);
  pinMode(ledPin, OUTPUT);

  lcd.begin(20,4);
  lcd.setCursor(0, 0);
  lcd.print("Priject #16");
  lcd.setCursor(0, 1);
  lcd.print("Industrial Motor");
  lcd.setCursor(0, 2);
  lcd.print("Speed Controller");
  delay(3000);
  lcd.clear();

  // show students info
  lcd.setCursor(0, 0);
  lcd.print("S1: Nada Alsaqqa");
  lcd.setCursor(0, 1);
  lcd.print("ID: 220212836");
  lcd.setCursor(0, 2);
  lcd.print("S2: Haneen Awad");
  lcd.setCursor(0, 3);
  lcd.print("ID: 220212135");
```

```
  delay(3000);
  lcd.clear();
}

void loop() {
  int potValue = analogRead(potPin); // read potentiometer
  // Map potentiometer value to PWM range (0-255) for motor speed
  int speedPWM = map(potValue, 0, 1023, 0, 255);
  // Map potentiometer value to simulated RPM range (0-470) for
display
  int rpm = map(potValue, 0, 1023, 0, 470);
  // Send PWM signal to transistor to control motor speed
  analogWrite(motorPin, speedPWM);

  lcd.setCursor(0, 0);
  lcd.print("System Running ");

  // Row 2: Current RPM
  lcd.setCursor(0, 1);
  lcd.print("Speed: ");
  lcd.print(rpm);
  lcd.print(" RPM        ");

  // Row 3: LED indicator status
  lcd.setCursor(0, 2);
  if (rpm >= 230) {
    digitalWrite(ledPin, HIGH);
    lcd.print("LED: ON  (High Speed)");
  } else {
    digitalWrite(ledPin, LOW);
    lcd.print("LED: OFF (Low Speed) ");
  }

  delay(200);
}
```

7. Code Explanation

Library & Pin Setup

The LiquidCrystal library controls the LCD using pins 12, 11, 5, 4, 3, 2. Three constants define the potentiometer (A0), motor PWM output (D9), and LED (D13).

setup()

Configures output pins, initializes the 20x4 LCD, and shows two splash screens: the project title for 3 seconds, then the student names and IDs for 3 seconds.

Reading the Potentiometer

analogRead(A0) returns a value from 0 to 1023. This is mapped to two ranges: PWM (0–255) for motor control, and RPM (0–470) for display.

PWM Motor Control

analogWrite(D9, speedPWM) sends a PWM signal through R1 (220Ω) to the gate of the IRLZ44N MOSFET (Q1). As the PWM duty cycle increases, the MOSFET conducts more,

allowing more current from the 12V supply to flow through the motor, raising its speed. The 1N4007 flyback diode (D2) suppresses voltage spikes when the motor switches off.

LCD Display

Row 0 shows 'System Running'. Rows 1 and 2 show the current RPM. Trailing spaces clear leftover characters when the digit count decreases.

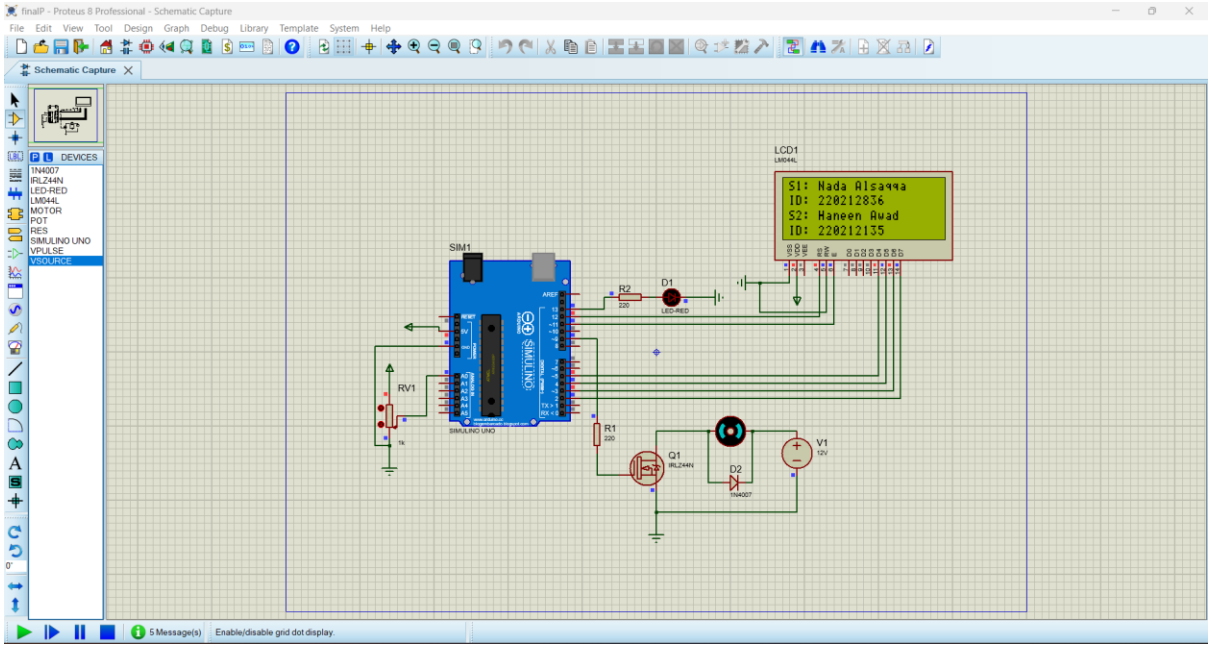
LED Warning

When RPM reaches 230 or above (about 49% of max), the LED turns ON and row 3 shows the high-speed message. Below 230 RPM the LED is OFF. The loop repeats every 200ms for smooth, flicker-free updates.

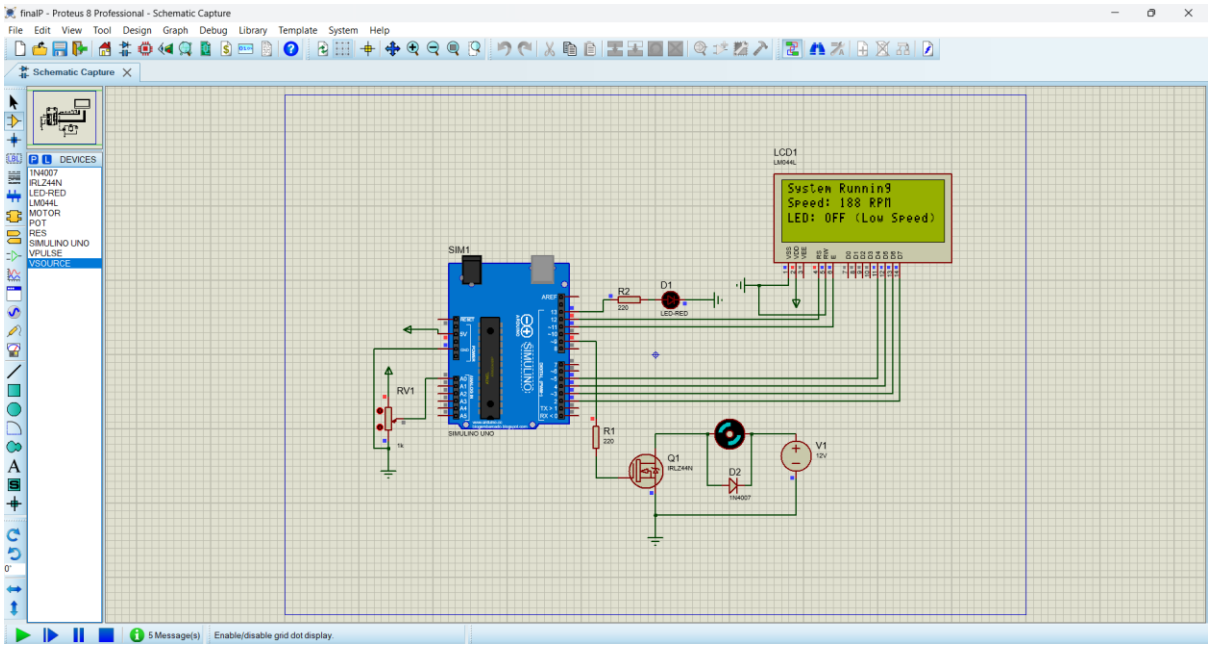
8. Simulation Screenshots

Insert your Proteus simulation screenshots below. Each box is a placeholder — paste or insert your screenshot image directly into it.

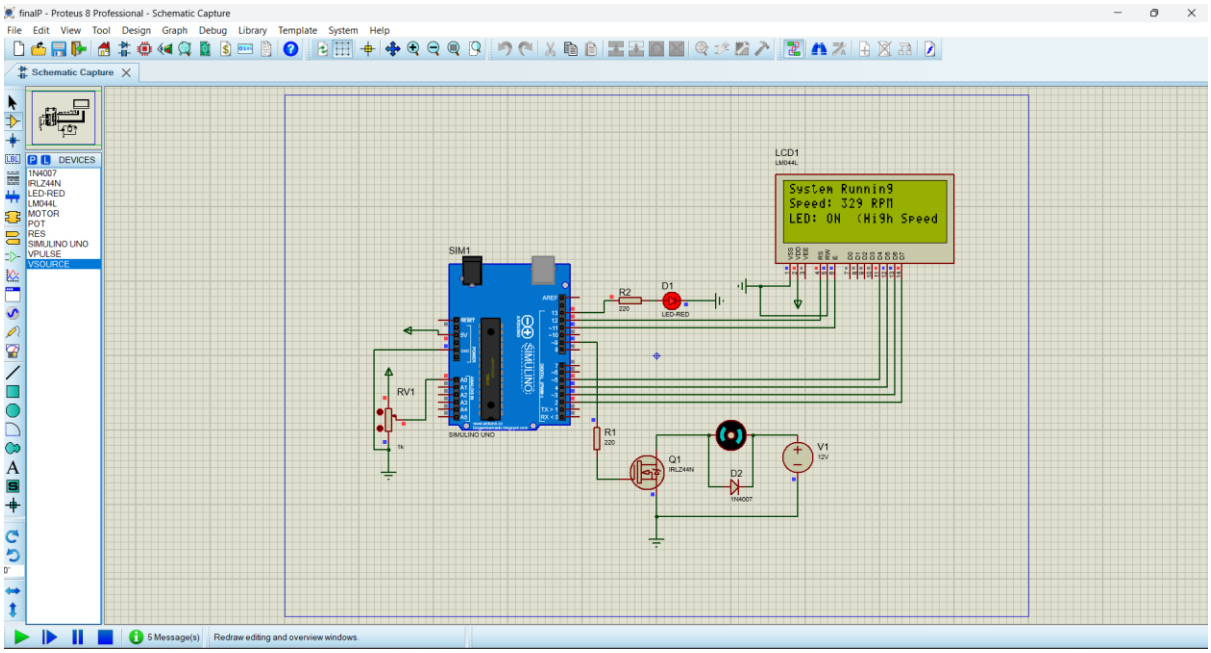
Screenshot 1: Startup – Student Info Screen



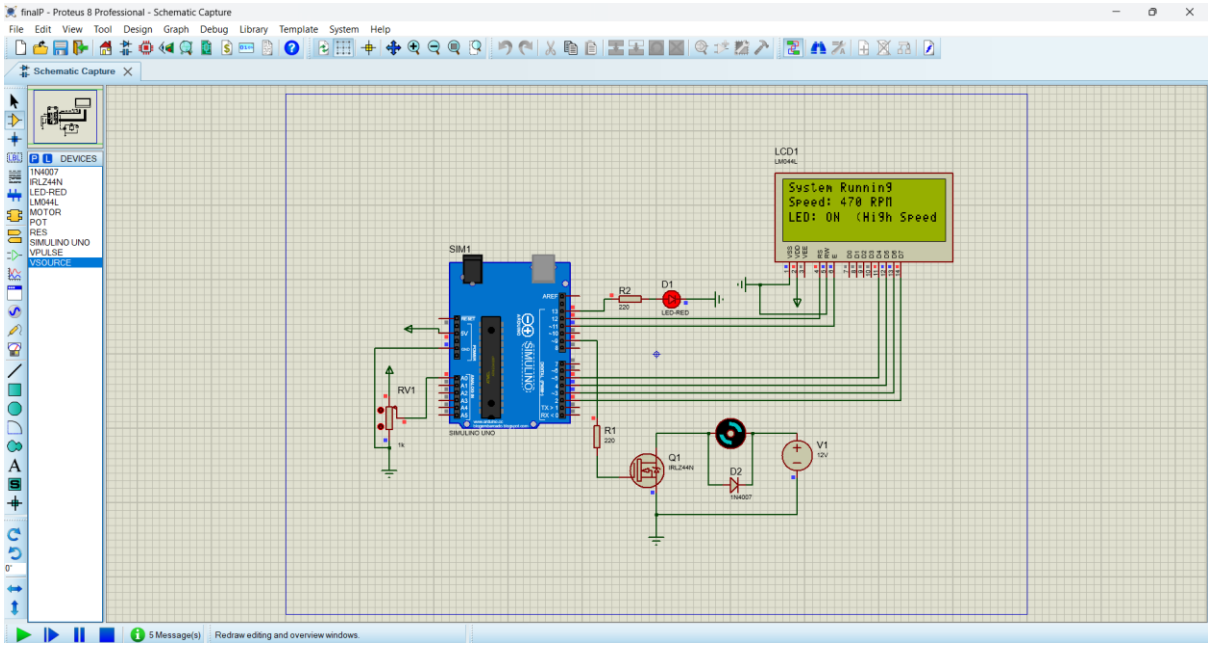
Screenshot 2: Low Speed Operation (RPM < 230)



Screenshot 3: High Speed Operation (RPM ≥ 230)



Screenshot 4: Maximum Speed (Potentiometer at 100%)



9. Conclusion

This project successfully demonstrates an Industrial Motor Speed Controller using Arduino UNO and Proteus simulation. It meets all project requirements by integrating a potentiometer (analog input), a DC motor with MOSFET driver (actuator), a 1N4007 flyback diode (protection), an LED (warning), and a 20x4 LM044L LCD (display).

The PWM approach provides smooth, continuous speed control from 0 to 470 RPM. The LED and LCD feedback give the operator clear real-time status. The project reinforces key embedded systems skills: ADC, PWM, MOSFET driving, LCD interfacing, and conditional control logic.